

T. Whirlybird Series. See *T. majus* Whirlybird Series.

TSUGA

PINACEAE

See also CONIFERS.

T. canadensis (Canada Hemlock, Eastern hemlock) illus. p.102. ‘**Aurea**’ is a broadly conical-shaped conifer, often with several stems. H 15ft (5m) or more, S 6–10ft (2–3m). Shoots are pale gray with spirally arranged, needlelike, flattened leaves, golden-yellow when young, aging to green in their second year, those along the top are inverted to show silver bands. Has ovoid, light brown cones. ‘**Bennett**’, H 3–6ft (1–2m), S 6ft (2m), is a compact, dwarf form with arching branches and a nest-shaped, central depression.

‘**Pendula**’ (syn. *T. c. f. pendula*) has weeping branches that may be trained to create a dome. H and S 10–15ft (3–5m); if left to spread at ground level, H 20in (50cm), S 6–15ft (2–5m). Z6–8

T. caroliniana (Carolina hemlock).

Conifer with a conical or ovoid crown. H 30–50ft (10–15m), S 15–25ft (5–8m). Red-brown shoots produce spirally set, needlelike, flattened, glossy, dark green leaves. Bears ovoid, green cones, ripening to brown. Z3–8

T. diversifolia (Japanese hemlock, Northern Japanese hemlock). Conifer with a broad, dense crown. H 30–50ft (10–15m), S 25–40ft (8–12m). Has orange shoots and needlelike, flattened, glossy, deep green leaves, banded with white beneath, that are spirally set. Ovoid cones are dark brown. Z6–8

T. heterophylla (Western hemlock). Vigorous, conical conifer with pendent branchlets. H 70–100ft (20–30m), S 25–30ft (8–10m). Gray shoots bear spirally set, needlelike, flattened, dark green leaves with silvery bands beneath. Green cones ripen dark brown. Z6–8

T. mertensiana (Mountain hemlock). Narrowly conical conifer with short, horizontal branches. H 25–50ft (8–15m), S 10–20ft (3–6m). Red-brown shoots bear needlelike, flattened, glaucous blue-green or gray-green leaves, spirally arranged. Cones are cylindrical and yellow-green to purple, ripening to dark brown. Z6–8. ‘**Glauca**’ illus. p.99.

T. sieboldii (Japanese hemlock, Southern Japanese hemlock). Broadly conical conifer. H 50ft (15m), S 25–30ft (8–10m). Glossy, buff shoots bear flattened, needlelike, lustrous, dark green leaves, set spirally. Cones are ovoid and brown. Z6–8

***Tsusiophyllum tanakae*.** See *Rhododendron tsusiophyllum*.

TUBERARIA

CISTACEAE

Genus of annuals and perennials, grown for their flowers. Fully to frost hardy. Requires sun and very well-drained soil. Propagate by seed in spring.

T. guttata, syn. *Helianthemum guttatum*. Moderately fast-growing, upright, branching annual. H and S 4–12in (10–30cm). Has lance-shaped, hairy, mid-green leaves. In summer, bears

yellow flowers, sometimes red-spotted at base of petals, that look like small, single roses. H5

TULBAGHIA

ALLIACEAE/LILIACEAE

Genus of clump-forming, mainly deciduous, sometimes semievergreen, rhizomatous or bulbous perennials. Grown for their umbels of dainty flowers, some fragrant, especially at night, which are borne over a long period from late spring to fall. Frost to frost tender. Needs full sun and well-drained soil. Propagate by division or seed in spring.

T. fragrans. See *T. simmleri*.

T. natalensis. Semievergreen, clump-forming perennial. H to 10in (25cm), S 4in (10cm). In midsummer, umbels of delicately fragrant, tubular, yellow-centered, white flowers, with spreading petal lobes, open above grasslike, mid-green foliage. Z10–11

T. pulchella. See *T. simmleri*.

T. simmleri, syn. *T. fragrans*,

T. pulchella, *T. simmieri*, illus. p.411.

T. violacea illus. p.280.

TULIPA

Tulip

LILIACEAE

Genus of mainly spring-flowering bulbs, grown for their bright, upward-facing flowers. Each bulb produces a few linear to lance-shaped, green or gray-green leaves on the stem. Flowers bear 6 usually pointed petals (botanically known as perianth segments) and 6 stamens, singly, unless otherwise stated below. Each plant has a spread of up to 8in (20cm). All are fully hardy unless otherwise stated. Requires a sunny position with well-drained soil and appreciates a summer baking; in cool, wet areas, bulbs may be lifted, when the leaves have died down, and stored in a dry place for replanting in fall. Propagate by division of bulbs in fall or, for species, by seed in spring or fall. ☉ If ingested, all parts may cause mild stomach upset, and contact with any part may aggravate skin allergies. See also feature panel pp.400–401. Horticulturally, tulips are grouped into the following divisions.

Single Early Group (Div. 1)—bear cup-shaped, single, white to dark purple flowers, to 3in (7cm) across, often margined, “flamed” or flecked with a contrasting color, from early to mid-spring. H 6–18in (15–45cm).

Double Early Group (Div. 2)—has bowl-shaped, fully double, dark red to yellow or white flowers, to 3in (8cm) across, often margined or flecked with another color; borne in mid-spring. H 12–16in (30–40cm).

Triumph Group (Div. 3)—sturdy stems bear cup-shaped, single flowers, to 2½in (6cm) across, in a range of colors and often margined or flecked with a contrasting color, in mid- and late spring. H 35–60cm (14–24in).

Darwin Hybrid Group (Div. 4)—has egg-shaped, single flowers, to 3in (7cm) across, in a range of colors and usually

flushed, “flamed” or margined with a different color and often with contrasting bases, from mid- to late spring. H 20–28in (50–70cm).

Single Late Group including Cottage and Darwin Hybrids (Div. 5)—has cup- or goblet-shaped, single flowers, sometimes several to a stem, in white to yellow, pink, red or almost black, often with contrasting margins, in late spring. H 18–30in (45–75cm).

Lily-flowered Group (Div. 6)—strong stems bear goblet-shaped, single flowers, to 3in (8cm) across, with reflexed, pointed petal tips, and sometimes margined, “flamed” or flushed with a contrasting color, in late spring. H 18–26in (45–65cm).

Fringed Group (Div. 7)—flowers are similar to those in Div. 6, but have fringed petals. H 14–26in (35–65cm).

Viridiflora Group (Div. 8)—has cup- or almost closed bowl-shaped, single flowers, to 3in (8cm) across, sometimes entirely green, margined with another color, or white to yellow, red or purple, “flamed” or striped green, with contrasting centers, borne in late spring. H 16–22in (40–55cm).

Rembrandt Group (Div. 9)—comprises mostly very old cultivars, similar to Div. 6, but has colors “broken” into striped or feathered patterns owing to virus. Flowers in late spring. H 18–26in (45–65cm).

Parrot Group (Div. 10)—bears cup-shaped, single, white to pink or violet-blue flowers, to 4in (10cm) across, often unevenly striped with different colors, including green, borne in late spring. Petals are finely and irregularly cut. H 14–26in (35–65cm).

Double Late Group (peony-flowered) (Div. 11)—has bowl-shaped, fully double flowers, to 5in (12cm) across, in white to purple, sometimes margined or “flamed” in a different color, borne in late spring. H 14–24in (35–60cm).

Kaufmanniana Group (Div. 12)—comprises *T. kaufmanniana* and hybrids and has bowl-shaped, single flowers, 3–4in (8–10cm) across, frequently multicolored and usually with distinctively colored bases, in early or mid-spring. Leaves are sometimes marked bronze, red, or purple. H 6–12in (15–30cm).

Fosteriana Group (Div. 13)—comprises *T. fosteriana* and hybrids and has bowl-shaped, single flowers, to 5in (12cm) across, in white to yellow or dark red, often margined or “flamed” in another color and with contrasting bases, borne in mid-spring. Leaves are sometimes marked red-purple. H 8–26in (20–65cm).

Greigii Group (Div. 14)—comprises *T. greigii* and hybrids and has bowl-shaped, single, yellow to red flowers, to 4in (10cm) across, sometimes “flamed” or margined in a different color and with contrasting bases, borne in early or mid-

spring. Blue-green leaves are generally wavy-margined and always marked dark bluish-maroon. H 6–12in (15–30cm).

Miscellaneous Group (Div. 15)—comprises all species and hybrids not included in other divisions. Flowers are produced from late winter to late spring.

T. ‘Abu Hassan’ (illus. p.401), Div. 3. Mid- to late-flowering spring bulb. H 20in (50cm). Produces cardinal-red flowers with pinkish-red stripes and yellow edges. Z3–8

T. acuminata (Horned tulip; illus. p.401), Div. 15. Mid-spring-flowering bulb. H 1–1½ft (30–45cm). Flowers are 3–5in (7–13cm) long, with long-pointed, tapered, pale red or yellow petals, often tinged with red or green outside. Z3–8

T. ‘Ad Rem’, Div. 4. Mid- to late spring-flowering bulb. H 2ft (60cm). Flowers are scarlet with black bases and yellow margins. Anthers are yellow. Z3–8

T. aitchisonii. See *T. clusiana*.

T. ‘Albert Heijn’, Div. 13. (illus. p.400). Early spring-flowering bulb. H 14in (35cm). Bears deep rose-pink flowers with a purple sheen and paler pink edges. Z3–8

T. ‘Ancilla’, Div. 12. Early spring-flowering bulb. H 6in (15cm). Flowers are pink and reddish outside, white inside, each with a central, red ring. Z3–8

T. ‘Angélique’, Div. 11. Late spring-flowering bulb. H 16in (40cm). Delicately scented, double, pale pink flowers deepen with age. Each petal has paler streaks and a lighter margin. Good for bedding. Z3–8

T. ‘Apeldoorn’s Elite’ (illus. p.401), Div. 4. Mid- to late spring-flowering bulb. H 2ft (60cm). Has buttercup-yellow flowers feathered with cherry-red and with yellowish-green bases. Z3–8

T. ‘Apricot Beauty’, Div. 1. Early spring-flowering bulb. H 16in (40cm). Flowers are salmon-pink, faintly tinged red. Z3–8

T. ‘Artist’ (illus. p.401), Div. 8. Late spring-flowering bulb. H 1½ft (45cm). Flowers are salmon-pink and purple outside, sometimes marked green, and deep salmon-pink and green inside. Z3–8

T. ‘Attila’, Div. 3. Mid-spring-flowering bulb. H 16in (40cm). Strong stems bear long-lasting, pink flowers. Z3–8

T. aucheriana, Div. 15. Early spring-flowering bulb. H to 8in (20cm). Has gray-green leaves. Bears yellow-centered, pink flowers, 2–5cm (¾–2in) long, each tapered at the base, and with ovate petals. Z3–8

T. australis. See *T. sylvestris*.

T. bakeri. See *T. saxatilis* Bakeri Group.

T. ‘Balalaika’, Div. 5. Late spring-flowering bulb. H 20in (50cm). Bright red flowers each have a yellow base and black stamens. Z3–8

T. ‘Ballade’ (illus. p.400), Div. 6. Late spring-flowering bulb. H 20in (50cm). Reddish-magenta flowers with a white-margined, yellow base have long petals that are margined white. Z3–8

T. ‘Ballerina’ (illus. p.401), Div.6. Late-flowering spring bulb. H 2ft (60cm). Produces lemon-yellow flowers with blood-red, flamelike markings, orange-yellow veins near the margins, and star-shaped yellow bases. Inner surfaces are bright red, feathered marigold-orange, with pale golden-yellow anthers. Z3–8

T. batalinii. See *T. linifolia*

Batalinii Group.